

Caifen Liu

PhD Candidate in Epidemiology and Biostatistics

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EDUCATION

The University of Hong Kong

Hong Kong, China

PhD in Epidemiology and Biostatistics

Sep 2022–Sep 2026 (expected)

Thesis: *Estimation and Interpretation of COVID-19 Severity and Vaccine Effectiveness.*

Supervisor: [Prof. Ben Cowling](#).

Southern University of Science and Technology

Shenzhen, China

MSc in Statistics

Sep 2018–Jun 2020

Thesis: *Bivariate Binomial Distribution for Comparing Binomial Observations in Two Correlated Groups.*

Supervisor: [Prof. Guo-Liang Tian](#).

Hunan University

Changsha, China

BE in Materials Science and Engineering

Sep 2014–Jun 2018

RESEARCH INTERESTS

I am interested in developing and applying causal inference methods to observational health data to reduce bias, strengthen the validity of causal conclusions, and support clinical decision-making.

Methodology: causal inference; causal estimands; selection bias; inverse-probability weighting; survival and longitudinal data analysis; simulation and sensitivity analyses; missing data methods; target trial emulation.

Applications: observational studies using linked electronic health records, registries, and cohort data; vaccine effectiveness evaluation; disease severity evaluation; pharmacoepidemiology.

DOCTORAL RESEARCH

The University of Hong Kong *School of Public Health*

2022–Present

COVID-19 severity pyramid and in-hospital progression

Investigated how changing admission practices altered COVID-19 severity estimates and longitudinal in-hospital progression. Modeled about 2.8 million linked cases, including about 120,000 hospitalizations, using Bayesian models of the severity pyramid, Aalen–Johansen estimators, and Cox models.

Hospital pressure, admission selection, and fatality risk

Assessed how hospital pressure, admission selection, and admission severity shaped the observed in-hospital fatality. Applied period-specific generalized additive models and model-based counterfactual standardization to about 120,000 hospitalizations.

Causal estimands and selection bias in test-negative designs

Examined when vaccine-induced severity attenuation drives test-negative estimates away from source-population causal estimands. Used causal DAGs, analytic bias derivations, Monte Carlo simulation, and inverse-probability weighting to evaluate bias and correction strategies.

Hospital-based test-negative design estimates of vaccine effectiveness (*ongoing*)

Uses simulation to evaluate when hospital-based TNDs recover vaccine effectiveness against target-pathogen hospitalization, benchmarked against randomized-trial and adjusted-cohort estimates under alternative hospital eligibility, testing, and control-selection mechanisms.

Peer-Reviewed Publications

†Co-first author

1. **Liu, C.**, Okoli, G. N., Chen, R., Sullivan, S. G., and Cowling, B. J. (2026). SARS-CoV-2 vaccination and attenuation of breakthrough infection severity: A systematic global review and meta-analysis. *Clinical Infectious Diseases*, ciag346.
2. Du, Z., **Liu, C.**[†], Bai, Y., Wang, L., Lim, W. W., Lau, E. H. Y., and Cowling, B. J. (2024). Predicting efficacies of fractional doses of vaccines by using neutralizing antibody levels. *JMIR Public Health and Surveillance*, 10, e49812.
3. Du, Z., **Liu, C.**[†], Wang, C., Xu, L., Xu, M., Wang, L., Bai, Y., Xu, X., Lau, E. H. Y., Wu, P., and Cowling, B. J. (2022). Reproduction numbers of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) variants: A systematic review and meta-analysis. *Clinical Infectious Diseases*, 75(1), e293–e295.
4. Du, Z., Wang, C., **Liu, C.**[†], Bai, Y., Pei, S., Adam, D. C., Wang, L., Wu, P., Lau, E. H. Y., and Cowling, B. J. (2022). Systematic review and meta-analyses of superspreading of SARS-CoV-2 infections. *Transboundary and Emerging Diseases*, 69(5).
5. **Liu, C.**[†], Xu, L., Bai, Y., Xu, X., Lau, E. H. Y., Cowling, B. J., and Du, Z. (2022). Local surveillance of the COVID-19 outbreak. *Frontiers in Physics*, 10, 824369.

Working Papers

1. **Liu, C.**, Murphy, C., Okoli, G. N., Sullivan, S. G., and Cowling, B. J. *Vaccine-induced severity attenuation and bias in test-negative estimates of vaccine effectiveness*. Manuscript in preparation.
2. **Liu, C.**, Cheung, J. K., Lin, Y., Wu, P., Wong, J. Y., and Cowling, B. J. *Hospital pressure, admission selection, and COVID-19 hospital fatality risk across epidemic periods in Hong Kong: a retrospective cohort study*. Manuscript in preparation.

SELECTED PRESENTATIONS

EPIDEMICS¹⁰

San Diego, USA

Oral presentation

Dec 2025

Impact of vaccine-induced severity attenuation on bias in vaccine-effectiveness estimates from test-negative designs.

EPIDEMICS⁹

Bologna, Italy

Poster presentation

Nov 2023

Estimating the COVID-19 severity pyramid in Hong Kong, January 2020–January 2023.

HKCCM Annual Scientific Meeting

Hong Kong, China

Poster presentation

Sep 2022

Surveillance for local and importation risk of COVID-19.

RESEARCH EXPERIENCE

Laboratory of Data Discovery for Health (D²4H)

Hong Kong, China

Research Associate

Jul 2021–Jul 2022

- Supported data access, protocol development, and ethics approval for a Hospital Authority Data Collaboration Lab retrospective cohort study of oseltamivir effectiveness in hospitalized influenza patients using longitudinal clinical records.

- Co-developed an R package integrating EpiEstim and virosolver to simulate epidemic trajectories, estimate time-varying reproduction numbers, and support real-time monitoring of local COVID-19 transmission.
- Conducted systematic reviews and meta-analyses of SARS-CoV-2 transmissibility, synthesizing variant-specific reproduction numbers and superspreading parameters.

Southern University of Science and Technology
Research Assistant, Department of Statistics and Data Science

Shenzhen, China
Jul 2020–Jun 2021

- Conducted methodological literature reviews and developed estimation methods using the MM algorithm for bivariate binomial distributions and multivariate zero-inflated count data.

TEACHING EXPERIENCE

The University of Hong Kong
CMED 6020 Advanced Statistical Methods I, *Tutor*
CMED 6020 Advanced Statistical Methods I, *Teaching Assistant*

Spring 2025–Spring 2026
Spring 2023–Spring 2024

Southern University of Science and Technology
MA103A Linear Algebra A, *Teaching Assistant*
MA102B Calculus A II, *Teaching Assistant*
MA101B Calculus A I, *Teaching Assistant*

Fall 2019
Spring 2019
Fall 2018

ADDITIONAL TRAINING

Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID)
University of Washington, Department of Biostatistics

July 2023

Selected modules: MCMC I and II for Infectious Diseases; Simulation-Based Inference for Epidemiological Dynamics; Statistics and Modeling with Novel Data Streams.

TECHNICAL SKILLS

Programming: R, Python; **Tools:** L^AT_EX, Git.

LANGUAGES

Mandarin (native); English (professional); Cantonese (conversational).